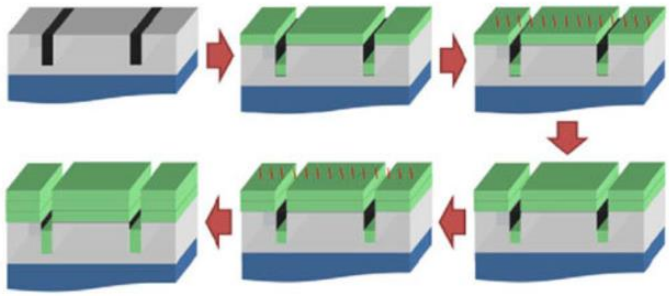
EXPERIMENT 4

Simulation-Based Analysis of Thin-Film Structure and Deposition Characteristics Using NanoHUB

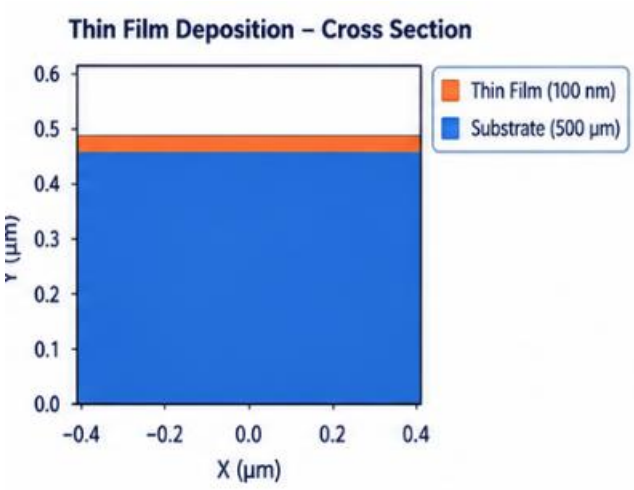
Circuit diagram:

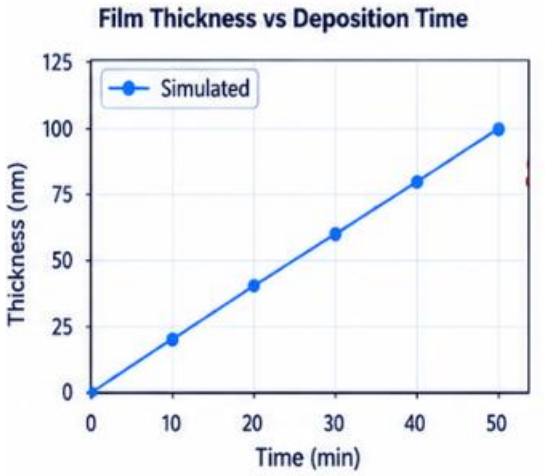


Output Parameter Table:

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	Deposition process	Proper thin film formation	Thin film deposited on the substrate	0% deviation
2	Thin film thickness	100 nm	100 nm	0% deviation
3	Substrate thickness	500 µm	500 µm	0% deviation
4	Deposition temperature	Selected value	300°C	0% deviation
5	Thin film structure	Substrate / Thin Film	Substrate / Thin Film	0% deviation
6	Film uniformity	Uniform layer	Uniform film layer observed	0% deviation
7	Film deposition rate	Calculated value	10 nm/min	0% deviation
8	Film/material quality	Proper deposited layer	Proper thin film layer observed	0% deviation

Output:


Thin Film Deposition – Cross Section
Thin Film (100 nm)
Substrate (500 µm)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Y (µm)
−0.4
−0.2
0.0
0.2
0.4
X (µm)


Film Thickness vs Deposition Time
Simulated
125
100
75
50
25
0
Thickness (nm)
0
10
20
30
40
50
Time (min)